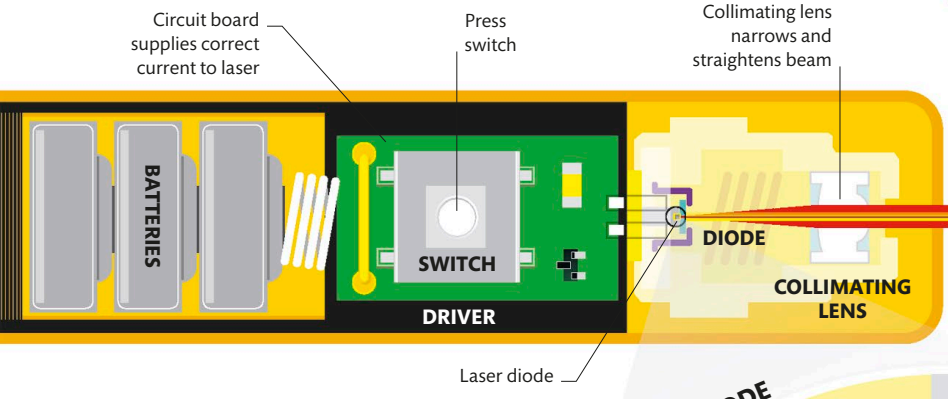
Lasers

A laser produces an intense beam of light that is collimated (all directed in a straight line, rather than spreading out) and coherent (all the waves are in step and all of the same frequency). The word "laser" stands for "light amplification by stimulated emission of radiation."

ARE LASERS USED AS WEAPONS?

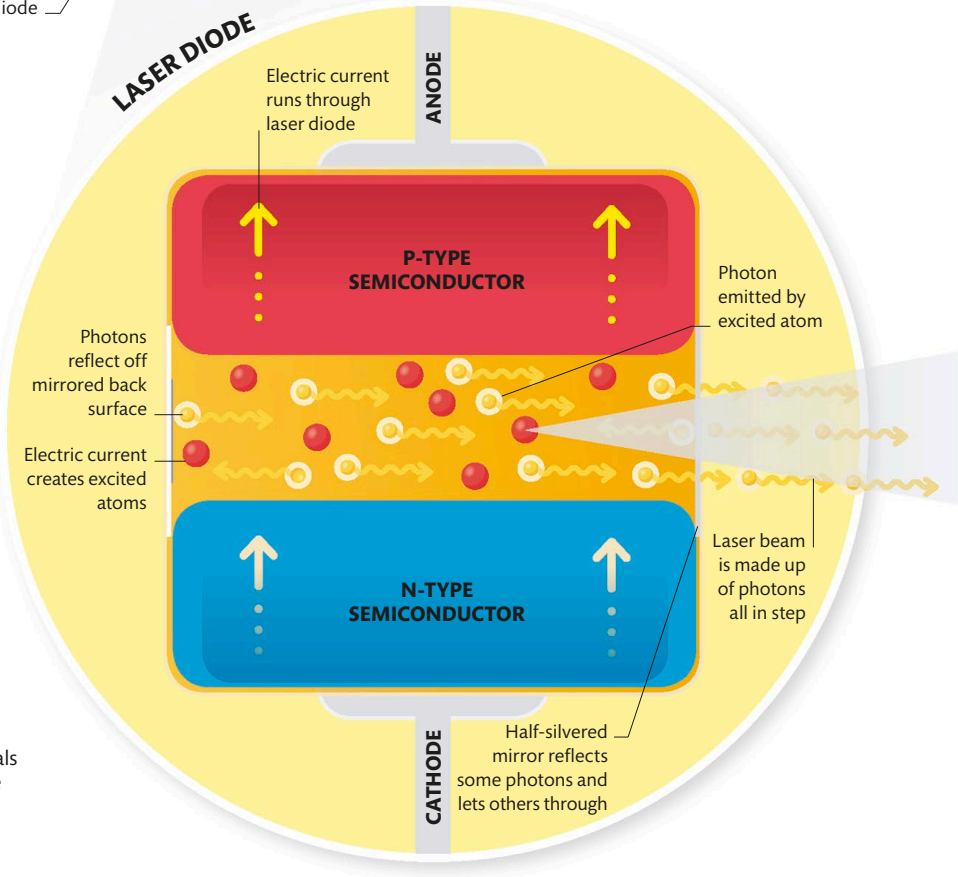
Yes, there are a few systems already in use, in which high-power lasers are used to destroy targets. At present, though, most systems are still experimental.



Laser pointer
People use lasers to highlight things on a projected slide. Inside a pointer are a diode (see below), battery, and electronic circuits.

Solid-state lasers

The most common lasers are low-power "solid-state" laser diodes, in which the light is produced by a solid sandwich of semiconductor materials. The outer layers are made of silicon combined, or "doped," with other elements to conduct electricity, while the inner layers are undoped. When an electric current flows through the layers, it initiates a process that leads to the production of a burst of light, called a photon (see opposite). Laser diodes are used in devices such as fiber-optic cables, laser printers, and barcode readers.



Laser diode
The two outer layers of semiconductor materials are "doped" n-type and p-type (see p.160). The center of the sandwich is undoped.